

Smart Hostel Maintenance Reporting System

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Agenda

- Introduction
- Aim and Objectives
- Questionnaire and Interview Design
- Data Collection and Target User Study
- Requirement Analysis
- Mental Model Representation (Hierarchical Task Analysis)
- Wireframing and Planning

Introduction

- Hostel maintenance issues (electrical faults, water leaks, broken furniture, etc.) are currently reported manually in person at the hostel office, through the Warden, or informally over WhatsApp.
- This manual process leaves students with no digital record, no way to track progress, and frequent delays or lost complaints, as identified in the Lab 1 target-user interviews.
- This project follows the Human Computer Interaction design process to design a Smart Hostel Maintenance Reporting System: a mobile-first application that lets students report issues quickly and track them through to resolution.
- Design decisions are grounded in real data a questionnaire completed by 122 hostel students across the Engineering, Technology and Agriculture faculties.
- This mid-presentation covers the stages completed so far: requirement analysis, task modelling (mental model / HTA), and interface planning (wireframes, navigation and system architecture).

Aim and Objectives

Aim

To design a user-centred digital system that allows hostel students to report and track maintenance issues quickly, transparently, and with minimal effort, replacing the current manual, untracked process.

Objectives

- Identify the target users and gather requirements through structured questionnaires and interviews (n = 122)
- Analyse and prioritise functional and non-functional requirements based on quantified pain points
- Model the student's mental model of the task using a Hierarchical Task Analysis (HTA)
- Design wireframes for all core screens covering the Access → Report → Track → Feedback cycle
- Plan the navigation flow and back-end system architecture needed to support the identified requirements
- Validate every design decision against the questionnaire data to ensure traceability from findings to design

1. Questionnaire / Interview Design

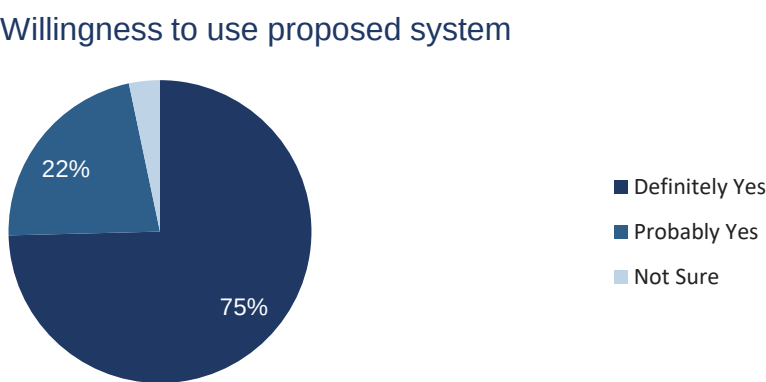
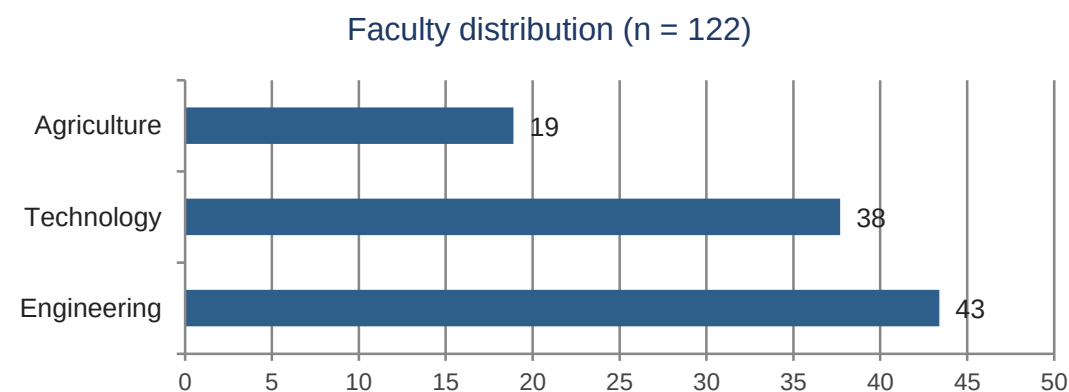
- Purpose: quantify the problems and preferences identified during the Lab 1 target-user interviews
- Instrument: Google Forms questionnaire distributed to undergraduate students living in the university hostels
- 122 responses collected across Engineering, Technology and Agriculture faculties, 1st-4th year students
- Structure: current reporting process, pain points, expectations for a digital system, feature preferences
- Question types: multiple-choice, Likert scale (1-5), multi-select, and open-ended free-text

2. Data Collection and Target User Study

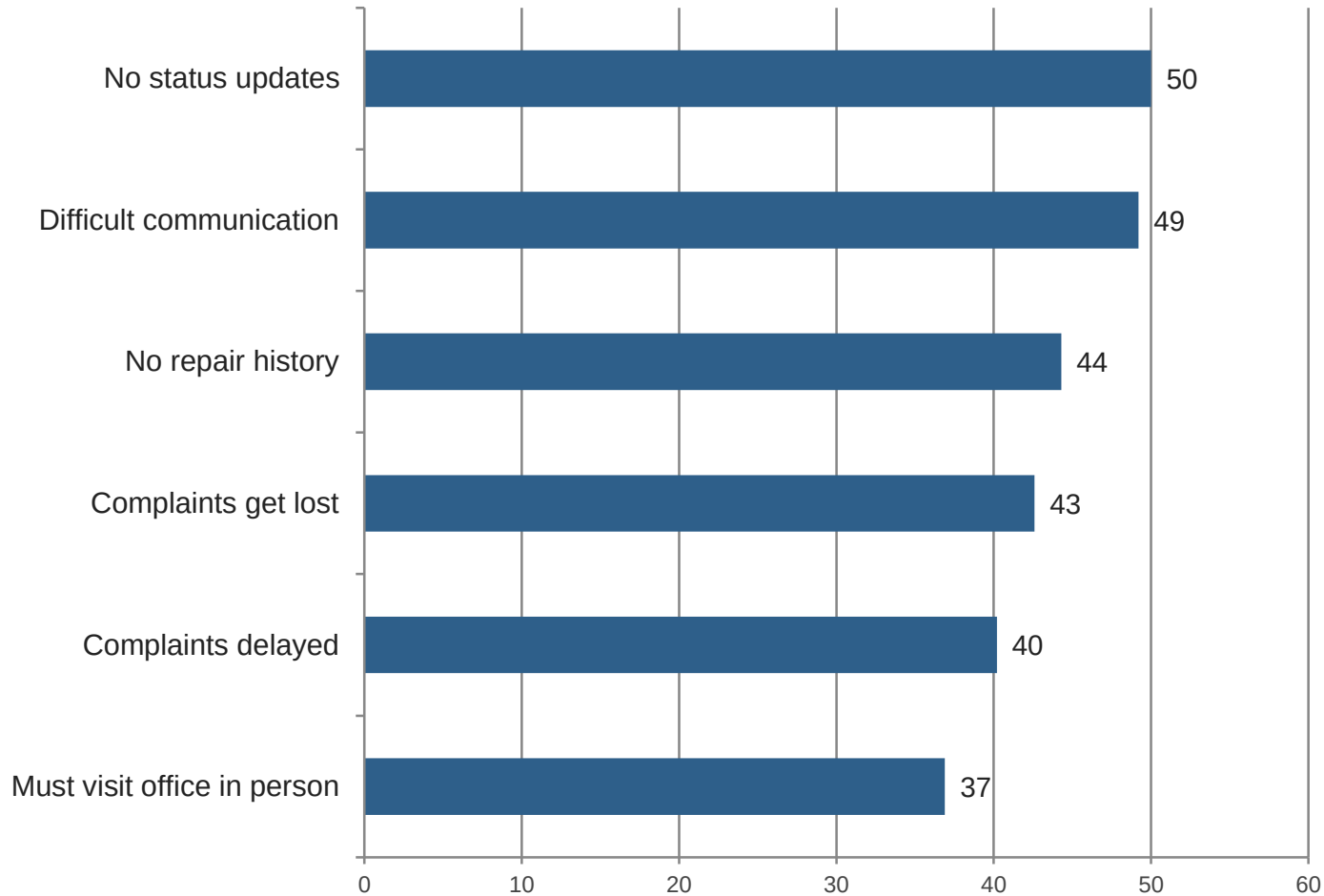
Primary user: Hostel student initiates the core task and was directly surveyed

Secondary users: Warden, Sub-Warden, Security, Maintenance Staff (administrative/operational roles from Lab 1)

Attribute	Distribution
Hostel	Boys' 53.3% Girls' 46.7%
Faculty	Engineering 43.4% Technology 37.7% Agriculture 18.9%
Year of study	1st 41.8% 4th 40.2% 3rd 16.4% 2nd 1.6%
Currently residing	Yes 95.9% No 4.1%



2. Key Findings -Problems with the Current Manual Process



- 89.3% have already reported an issue manually
- 75.4% avoided reporting an issue at least once because the process was difficult
- Average overall satisfaction with current process: 2.22 / 5
- 50.0% never know their complaint's status after reporting

3. Requirement Analysis

Goal 0: *Successfully report a hostel maintenance issue and track it through to resolution with minimal effort and full visibility of progress.*

Functional Requirements (highlights)

- Submit a complaint in 1-2 steps with optional photo/video evidence
- Real-time status tracking (Submitted → Assigned → In Progress → Resolved)
- QR-code room scan to auto-fill location
- Emergency flag and anonymous reporting options
- In-app chat with assigned maintenance staff
- Post-repair rating and feedback; per-room complaint history

Non-Functional (Usability) Requirements

- Very Simple layout by default, large labelled buttons (4.52/5 importance)
- Both Dark Mode and Light Mode supported
- Sinhala, Tamil and English language support
- Mobile-optimised first (82% primary device); usable on desktop for admins
- Minimise steps/fields to reduce abandonment (75.4% avoided reporting)
- Continuous, visible complaint status at every stage

4. Mental Model Representation – Hierarchical Task Analysis

The HTA represents the student's mental model: the sequence they expect to follow to resolve an issue, and the visibility they expect at each stage.

0. Report and track a hostel maintenance issue through to resolution

Plan 0: Do 1 → 2 → 3 in sequence; repeat within 3 until status = Resolved; then do 4.

Sub-task	Actions / Operations	Implemented on screen
1. Access the reporting system	Open the app · Log in with student credentials	Login / Access
2. Submit a maintenance complaint	Identify location · Select category · Attach evidence (optional) · Flag emergency/anonymous (optional) · Submit	Report Issue
3. Track the complaint until resolved	Open My Complaints · View status and ETA · Receive notifications · Use chat if needed	My Complaints, Complaint Detail
4. Provide post-repair feedback	Confirm resolution · Rate staff and repair speed · Add optional comment	Feedback

5. Navigation and System Architecture

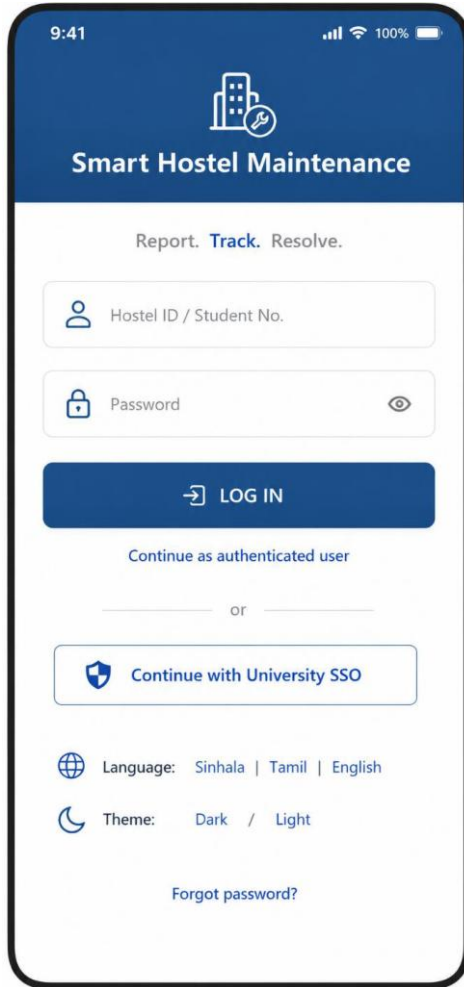
Navigation flow

- Login / Access → Home
- Home → Report Issue → My Complaints
- My Complaints → Complaint Detail
- Complaint Detail → Feedback (once Resolved)
- Bottom navigation bar (persistent): Home · Report · My Complaints · Profile
- Push notifications open the Complaint Detail screen directly

Back-end architecture

Layer	Components
Front-end	Student Mobile App (primary), Admin Web Dashboard, Security/Sub-Warden Portal
API Gateway	Single REST/GraphQL gateway – authentication and routing
Back-end services	Complaint Service, User & Auth, Notification Service, Chat Service
Data & storage	Relational database, cloud media storage, notification queue

5.1 Login / Access Screen



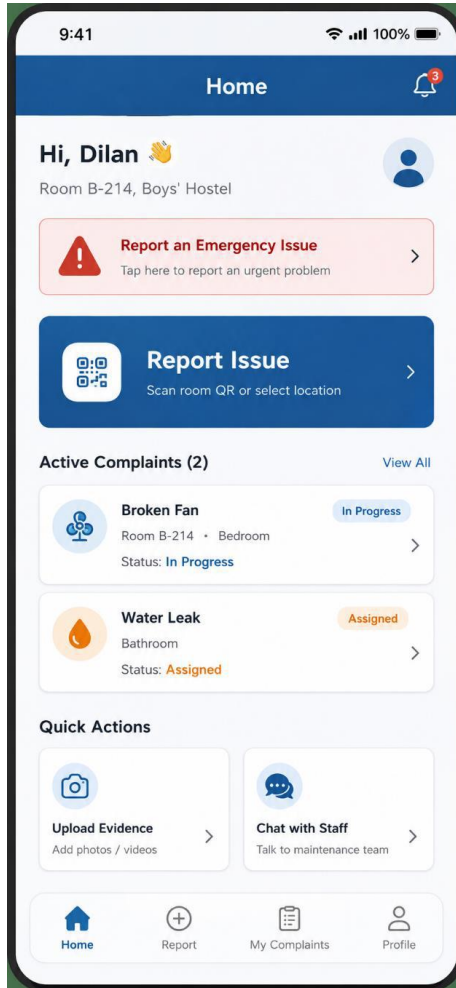
HTA Sub-task 1 – Access the reporting system

Supports hostel ID login or continuing an authenticated session, plus language and theme selection up front so the app is usable immediately after opening.

Key GUI elements

- Hostel ID / Student No. field and masked Password field
- LOG IN primary button; Continue as authenticated user link
- Continue with University SSO (alternate authentication)
- Language selector (Sinhala / Tamil / English)
- Theme toggle (Dark / Light); Forgot password? link

5.2 Home Screen



Entry point after HTA Sub-task 1, leading into Sub-task 2

The landing screen after login. Surfaces the emergency-report shortcut (a top-requested feature, 40.2%) and the primary Report Issue action, with a preview of active complaints so students see status without extra navigation.

Key GUI elements

- Greeting with student name, room and hostel
- Report an Emergency Issue banner (priority shortcut)
- Report Issue primary button (large, high contrast)
- Active Complaints preview cards with status
- Quick actions: Upload Evidence, Chat with Staff
- Persistent bottom navigation: Home / Report / My Complaints / Profile

5.3 Report Issue Screen

9:41 100%

Report Issue

1 Step 1 of 2 2

Scan Room QR Code
Use camera to auto-fill room details

or select manually

Hostel
Boys' Hostel

Room
B-214

Issue Category

Electrical Water Leak

Furniture Door Lock

Attach Evidence (optional)

Add Photo Add Video

☐ Flag as Emergency

☐ Submit Anonymously

Description (optional)

Add any extra details...

SUBMIT COMPLAINT

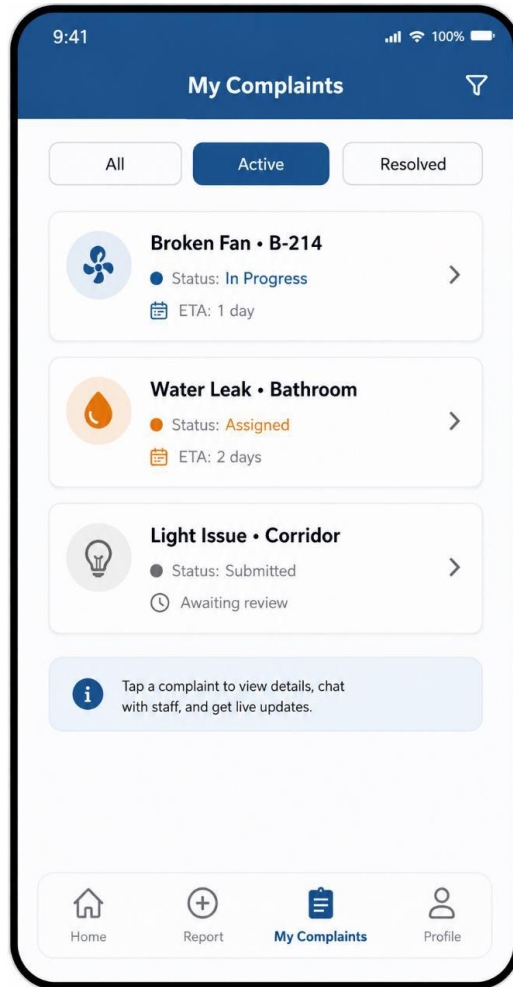
HTA Sub-task 2 – Submit a maintenance complaint

Designed to be completed in 1–2 steps (85.2% preference): QR scan or manual location selection, category selection, optional photo/video evidence, emergency/anonymous flags, and a single submit action.

Key GUI elements

- Scan Room QR Code button (auto-fills location)
- Hostel and Room dropdowns as a manual fallback
- Issue category chips: Electrical, Water Leak, Furniture, Door Lock
- Add Photo / Add Video buttons (optional evidence)
- Flag as Emergency and Submit Anonymously checkboxes
- Description text area and Submit Complaint button

5.4 My Complaints Screen



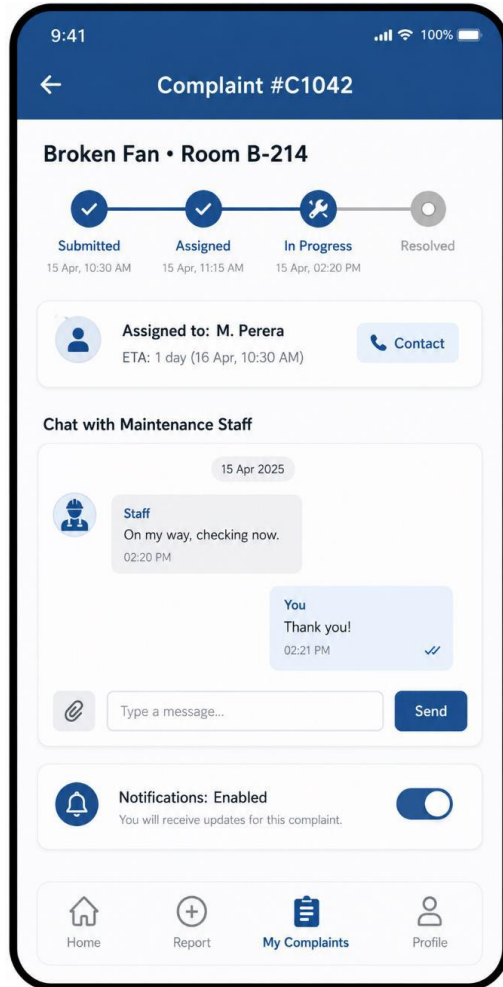
HTA Sub-task 3 – Track the complaint (start)

Implements the start of tracking. Filterable by All / Active / Resolved, with colour-coded status dots addressing the top pain point that 50.0% of students never know a complaint's status.

Key GUI elements

- Tab control: All / Active / Resolved
- Complaint summary cards: title, location, colour-coded status dot, ETA
- Tap a complaint to open its detail, chat, and live updates
- Shared bottom navigation bar

5.5 Complaint Detail Screen



HTA Sub-task 3 – Track the complaint (continued)

A horizontal progress tracker shows the four-stage status (Submitted, Assigned, In Progress, Resolved), alongside in-app chat with the assigned maintenance staff – the second most-requested feature (33.6%).

Key GUI elements

- Stepper / progress tracker: Submitted → Assigned → In Progress → Resolved
- Assigned staff name and ETA, with a Contact shortcut
- Chat window with message bubbles (staff / student)
- Text input and Send button to compose a chat message
- Notifications: Enabled status confirmation

5.6 Feedback Screen

HTA Sub-task 4 – Provide post-repair feedback

Appears once a complaint reaches Resolved status. Captures a staff rating and a repair-speed rating plus optional comments, then closes the complaint into the student's history.

Key GUI elements

- Issue Resolved confirmation banner
- Star rating control – Rate the Maintenance Staff
- Star rating control – Rate the Speed of Repair
- Additional Comments text area (optional)
- Submit Feedback button – closes complaint, updates history

Conclusion

Finding	Design Response
50% of students never see complaint status	Four-stage status tracker with push notifications
82% use mobile as primary device	Mobile-first app with optional web dashboard
85.2% want 1–2 step submission	QR auto-fill and minimal-field Report Issue form
Complaint tracking is the #1 requested feature	My Complaints and Complaint Detail screens

Next step: usability testing of these wireframes with hostel students, followed by detailed implementation.

Thank you